



Cost & Effort Analysis

From working prototype to a shippable marketplace app — build effort, 5-year mobile cost of ownership, and whether self-hosting on AWS is worth it at early scale.

Prepared for the SwapHub project · **July 2026**

Scope: early app, ~1,000 users, trade-only · FX: 1 USD ≈ **₱57**

Live prototype: **swaphub-web.vercel.app**

Executive summary

SwapHub is already a working, responsive web prototype (10 screens, deployed and publicly accessible). This document estimates what it takes to turn it into a real product for a first cohort of ~1,000 users, the 5-year cost of running it as a native mobile app, and how a self-hosted AWS setup compares to managed services at that scale.

₱400k–1.0M

Early-app build (1k users)

2–3 mo

To beta launch

< ₱2k/mo

Infra at 1k users (managed)

₱1.5M–3.2M

5-yr native app (contracted)

Key takeaways

- **The UI is done.** Remaining work is wiring real data and backend logic, not redesigning screens.
- **Go PWA-first or Expo.** One codebase covers web + iOS + Android; skip separate native builds to halve cost.
- **Managed services beat AWS at 1k users.** Supabase / Cloudinary / Vercel free tiers cost near ₱0; raw AWS carries fixed monthly baseline plus DevOps labour.
- **Maintenance is the real 5-year cost of native apps** — Apple/Google force yearly updates whether or not you add features.
- **Self-hosting only pays off at scale** (roughly 50k–100k+ users), not at launch.

Recommended path

Build a lean, trade-only app on **Supabase + Expo** reusing this UI, launch a private beta to ~1,000 users, stay on free/managed tiers, and revisit AWS only if you reach real scale.

1 Early app for the first 1,000 users

The smallest real version: sign up, post items with photos, search, chat, and confirm a trade. Payment-free to avoid escrow, KYC and financial licensing.

Build effort (lean scope)

Workstream	Approach	Person-wks	Est. cost
Foundations	Supabase project, data model, deploy pipeline	1.5	₱50k–80k
Auth & profiles	Supabase Auth (OTP), profile pages	2	₱70k–110k
Listings + photos	CRUD + Supabase Storage uploads	3	₱110k–160k
Search & filters	Postgres full-text + filters	1.5	₱50k–80k
Home feed	Recommended / near-you queries	1	₱35k–55k
Chat / messaging	Supabase Realtime threads + images	2.5	₱90k–130k
Offers & trade flow	Make offer, confirm, complete	2	₱70k–110k
Reviews & ratings	Post-trade reviews, reputation	1	₱35k–55k
Notifications	Email (Resend) + web push	1	₱35k–55k
Report + mini admin	Report button, simple admin list	1	₱35k–55k
Wire UI → real data	Connect existing screens (PWA shell)	3	₱110k–160k
QA & beta polish	Testing, bug fixing, launch prep	2	₱70k–110k
Total	~21.5 person-weeks	21.5	₱400k–1.0M

Solo builder ≈ 5 months part-time; two builders ≈ 2–3 months. Blended rate assumed ₱900–1,300/hour.

Ways to build v1

Option	Cost	Timeline	Notes
Solo / founder-built	~₱0 cash	4–6 months	Your time + this UI; only pay domain
You + 1 full-stack dev (recommended)	₱400k–800k	2–3 months	Best speed/cost balance
Mini-agency package	₱600k–1.2M	2–3 months	Least effort, more cash

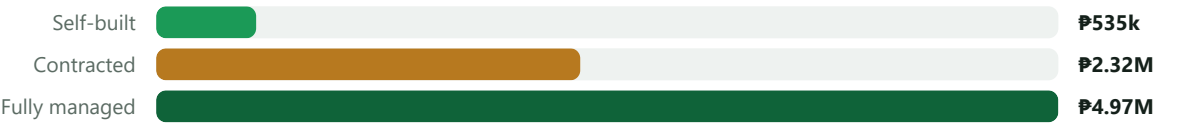
2 5-year cost as a native mobile app

Published to the App Store and Google Play, kept running for 5 years at ~1,000 users with minimal bug fixes. Assumes one shared Expo (React Native) codebase for both stores.

5-year total — three scenarios

Scenario	Build	Store+domain	Infra	Maintenance	5-yr total
Self-built & self-maintained	₱0–500k	₱35k	₱90–180k	your time	₱125k–715k
Contracted + light retainer	₱500k–1.2M	₱35k	₱90–180k	₱0.9–1.8M	₱1.5M–3.2M
Fully managed (agency SLA)	₱800k–1.5M	₱35k	₱120–200k	₱2.4–4.8M	₱3.4M–6.5M

5-year cost composition (midpoints, PHP thousands)



Bars scaled to the largest scenario. Maintenance is the dominant driver in the contracted and managed cases.

Why maintenance isn't optional for native apps

Even with zero new features, Apple and Google ship breaking OS updates yearly and change store policies. An app left untouched for 12–18 months typically breaks or gets delisted. Budget at least a light retainer (~₱15k–30k/mo) to stay installable and compliant. A single Expo codebase roughly halves this versus separate iOS + Android apps.

Recurring fees at ~1,000 users

Item	Rate	5-year cost
Apple Developer Program	\$99/yr (₱5,650)	₱28,250
Google Play (one-time)	\$25 (₱1,430)	₱1,430
Domain	~₱1,000/yr	₱5,000
Supabase Pro (DB/auth/storage/realtime)	~\$25/mo	₱85,500
Images (Cloudinary), email, push, monitoring	Free tiers	₱0

3 Self-hosting on AWS — does it lower cost?

Short answer: no, not at ~1,000 users. Raw AWS adds a fixed monthly baseline *and* ongoing DevOps labour. Managed/free tiers are near-zero at this scale. AWS only becomes cheaper per user at large scale (~50k–100k+ users) or when specific control/compliance is required.

Monthly infrastructure at 1k users: AWS vs managed

Component	Self-hosted AWS	Managed (Supabase etc.)
Compute (API server)	\$15–30 (EC2/Fargate)	included
Database (Postgres)	\$15–25 (RDS)	₱0–1.4k (Supabase)
Load balancer (ALB)	\$18–22 fixed	included
NAT gateway (if used)	~\$32 fixed	n/a
Object storage + CDN	\$3–15 (S3+CloudFront)	₱0 (Cloudinary free)
Data transfer, backups, CloudWatch	\$10–30	included
Monthly total	~\$60–150 (₱3.4k–8.6k)	₱0–2k

A bare single EC2 box (Postgres on the same instance, no ALB/NAT) can run ~\$15–35/mo, but you lose managed backups and high availability and take on *all* patching and ops yourself.

5-year comparison (including labour)

Cost bucket	Self-hosted AWS	Managed
Infrastructure (5 yr)	₱200k–500k	₱90k–200k
One-time setup (VPC, IaC, CI/CD, hardening)	₱80k–200k	~₱0 (in build)
DevOps maintenance (patching, scaling, backups)	₱600k–2.4M	~₱0 (managed)
5-year infra + ops	₱880k–3.1M	₱90k–200k

The hidden cost is people, not servers

The AWS bill itself isn't huge, but self-hosting means **someone must own security patching, database upgrades, backups, monitoring and scaling**. That is the opposite of "minimal bug fixes" — it adds a standing DevOps responsibility that managed platforms absorb for you.

When AWS *does* make sense

- You cross ~50k–100k+ active users and managed per-unit prices exceed AWS baseline + amortized DevOps.
- You need specific compliance, data residency, or custom infrastructure managed platforms can't offer.
- You already have in-house DevOps whose time is effectively "free" to the project.
- Heavy, predictable workloads (large media/video, ML) where reserved AWS capacity beats usage-priced SaaS.

Recommendation: Stay on managed services (Supabase + Cloudinary + Vercel/Expo) for the early app. Design the backend cleanly so a future migration to AWS is possible — but don't pay the AWS complexity tax until real scale justifies it.

4 Bottom line

₱400k–1.0M

Lean build (1k users)

₱1.5M–2.5M

5-yr native app
(recommended)

₱90k–200k

5-yr managed infra

₱880k–3.1M

5-yr AWS infra + ops

Recommended plan

- **Build:** Trade-only app on Supabase + Expo, reusing the existing UI. Target 2–3 months, ₱400k–800k with one contractor (or your own time).
- **Launch:** Private beta to ~1,000 invited users. PWA first if you want to validate before paying store fees; add native store builds once demand is proven.
- **Infrastructure:** Managed/free tiers. Expect < ₱2k/month at this size. Do not self-host on AWS yet.
- **Maintenance:** Light retainer (~₱15k–30k/mo) once on the stores, purely to keep up with OS and policy changes.
- **Revisit AWS** only after ~50k+ users or a specific compliance need.

Total to get live and run for a year

Build (₱400k–800k) + one year of managed infra (~₱20k) + store/domain fees (~₱8k) ≈ **₱430k–830k for year one**, or dramatically less if you build it yourself.

All figures are planning ranges for the Philippine market as of July 2026, not fixed quotes. Actual costs vary with team rates, feature scope, and usage. FX assumption 1 USD ≈ ₱57. Live prototype: swaphub-web.vercel.app